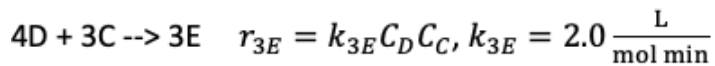
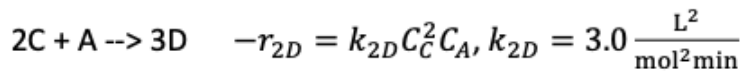
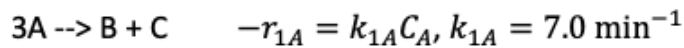


E08 Multiple Reactions in an Isothermal CSTR, 2 ways

K_{IA}^c

The following reactions are taking place in liquid phase in a CSTR:



The entering volumetric flowrate is 100 L/min and the entering concentration of A is 3 mol/L. The CSTR operates at $C_A = 0.10 \text{ M}$, $C_B = 0.93 \text{ M}$, $C_C = 0.51 \text{ M}$, $C_D = 0.049 \text{ M}$.

- 1) Determine the CSTR volume.

We get the volume from *the mole balances*

Here we have 5 species, we can use the mole balance for *any of them*

At first glance, species E might be a good choice, since *it is only involved in one reaction*

However, *we are not given its concentration so its mole balance doesn't give us enough information*

Looking again, species *B* only is involved in *one reaction*

Writing the net rate of species B and solving:

$$r_{B,net} = r_{1B}$$

$$\frac{r_{1B}}{1} = \frac{r_{1A}}{-3}, \text{ so } r_{1B} = -\frac{1}{3} r_{1A} = \frac{1}{3} k_{1A} C_A,$$

mole balance on B:

$$V = \frac{F_{B0} - F_B}{-r_{B,net}} \quad F_{B0} = 0, \quad F_B = C_B v_0,$$

$$-r_{B,net} = -\frac{7}{3 \text{ min}} (0.10 \frac{\text{mol}}{\text{L}})$$

$$\text{so } V = \frac{(100 \frac{\text{L}}{\text{min}}) (0.93 \frac{\text{mol}}{\text{L}})}{(\frac{7}{3 \text{ min}}) (0.10 \frac{\text{mol}}{\text{L}})} = 399 \text{ L}$$

- 2) Let's pretend we don't know any of the exit concentrations, but we know the volume is 400 L. Determine the exit concentrations.

Now, the number of unknowns = 5

We hence need 5 equations. The equations that we will use are the

mole balances

These take the form

$$V \frac{F_{j0} - F_j}{-r_{j,net}}$$

Alternatively, we can write them

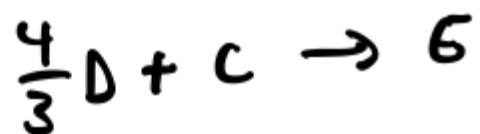
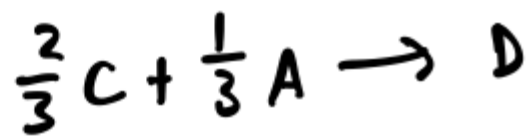
$$0 = F_{j0} - F_j + V r_{j,net}$$

Setting them equal to zero sets us up well to use a *non-linear* equation solver such as *fsolve*, which finds the "roots" of an equation

To do this, first let's write the

the stoichiometry matrix $\frac{1}{\nu}$, the rate matrix

$$r = \begin{bmatrix} k_{1A} C_A \\ k_{2D} C_C^2 C_A \\ k_{3E} C_D C_E \end{bmatrix}$$



Stoichiometry matrix:

$$\nu = \begin{bmatrix} -1 & \frac{1}{3} & \frac{1}{3} & 0 & 0 \\ -\frac{1}{3} & 0 & -\frac{2}{3} & 1 & 0 \\ 0 & 0 & -1 & -\frac{4}{3} & 1 \end{bmatrix}$$